

CSc 360

Operating Systems

Process Synchronization

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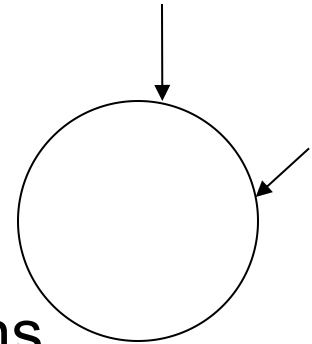
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The need for synchronization

- Multiprogramming
 - multi-process
 - process communication
 - shared memory or message passing
 - multi-thread
- CPU scheduling
- Cooperating processes/threads
 - e.g., the “producer-consumer” problem
 - cannot consume the things not produced yet

The producer-consumer problem

- Solutions so far
 - bounded buffer
 - in, out variables; full, empty conditions
 - N buffer space, N-1 utilized at most
 - first-in-first-out queue
 - FIFO variable
- A simpler solution
 - to fully utilize the circular buffer
 - use a “counter” variable



The “counter” solution

```
while (true) {  
    /* produce an item and put in nextProduced */  
    while (count == BUFFER_SIZE) ; // do nothing  
    buffer [in] = nextProduced;  
    in = (in + 1) % BUFFER_SIZE;  
    count++;  
}
```

Producer

```
while (true) {  
    while (count == 0) ; // do nothing  
    nextConsumed = buffer[out];  
    out = (out + 1) % BUFFER_SIZE;  
    count--;  
    /* consume the item in nextConsumed */  
}
```

Consumer

Race condition

- E.g., increment a counter (shared variable)
 - read the counter (from memory)
 - increment by one (at CPU)
 - write the counter
- How about two threads?
 - *sharing* only one counter e.g., counter=5 initially
 - non-deterministic result: $R_1W_1R_2W_2$; $R_1R_2W_1W_2$
- “There is something not to be (always) shared”

Critical section

- Critical section
 - code section accessing shared data
 - only one thread executing in critical section
 - only one thread accessing the shared data: serialize
 - choose the right (size of) critical section!
- Approach: exclusion (lock)
 - if locked, wait!
 - if not lock, lock (and later, unlock)

Properties of “solutions”

- Mutual exclusion
 - no more than one process in the critical section
- Making progress
 - if no process in the critical section, one can in
- Bounded waiting
 - for processes that want to get in the critical section, their waiting time is bounded

Problem formulation

- Only 2 processes, P_0 and P_1
- General structure of process P_i (other process P_j)
 do {
 entry section
 /* critical section */
 exit section
 /* remainder section */
 } **while** (1);
- Processes may **share** some common variables to synchronize their actions
 - do not get into the loop!

Algorithm 1

- Shared variables
 - `int turn; // initially turn = 0`
 - `turn == i`: P_i can enter its critical section
- Process P_i
 - `do {`
 - `while (turn != i) ; // wait`
 - `/* critical section */`
 - `turn = j;`
 - `/* remainder section */`
 - `} while (1);`
- Fate on other's hands: any problems?

Algorithm 2

- Shared variables
 - **boolean flag[2];**
initially **flag [0] = flag [1] = false.**
 - **flag [i] = true** : P_i ready to enter its critical section
- Process P_i
 - do {
 - flag[i] := true;**
 - while (flag[j]) ;** // wait
 - /* critical section */**
 - flag [i] = false;**
 - /* remainder section */**
 - } while (1);**
- Fight for access: any problems?



Dekker's solution

- Combined shared variables of Algorithms 1 and 2
- Process P_i

```
while (true) {  
    flag[i] = true;  
    while (flag[j]) {  
        if (turn == j) {  
            flag[i] = false;  
            while (turn == j) ; // wait  
            flag[i] = true;  
        }  
    }  
    /* critical section */  
    turn = j;  
    flag[i] = false;  
    /* remainder section */  
}
```

- Be polite: meet all three requirements; solve the critical-section problem for **two** processes

Peterson's solution

- A simpler solution
 - combined shared variables of Algorithms 1 and 2

- Process P_i

do {

 flag [i] := true;

 turn = j;

 while (flag [j] and turn == j) ; // wait

 /* critical section */

 flag [i] = false;

 /* remainder section */

 } while (1);

- Meet all three requirements; solve the critical-section problem for **two** processes

This lecture

- Process synchronization
 - the producer-consumer problem
 - software solution for 2 processes
 - Peterson's solution
- Explore further
 - Lamport's bakery algorithm
 - for n processes
 - it's time to google!

Next lecture

- Process synchronization
 - other alternatives (read OSC7Ch6)